

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Real-Time Scalable Plant Disease Classification Using EfficientNet

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Plant diseases are a major threat to global agriculture, causing an estimated 20–40% loss in crop production every year and resulting in financial damages exceeding \$220 billion annually. The problem is particularly severe for small-scale farmers who often lack access to timely expert diagnosis. Delays in identifying plant diseases can lead to irreversible damage, as infections spread rapidly and reduce the effectiveness of preventive measures.

Although deep learning techniques have shown promising results in plant disease detection, most existing models are trained on controlled datasets with uniform backgrounds and ideal lighting conditions, which do not reflect real-world farming environments. Additionally, many high-performing models are computationally expensive and unsuitable for deployment on affordable mobile devices. This research addresses the need for a practical, real-time solution by developing a scalable plant disease classification system using EfficientNet that balances accuracy with computational efficiency.

Despite significant advancements in deep learning for plant disease classification, there remains a critical gap between laboratory performance and real-world applicability. Most existing models achieve high accuracy on benchmark datasets such as PlantVillage but fail to maintain similar performance when tested on real-world field images due to variations in lighting, background noise, occlusion, and image quality. Furthermore, widely used architectures like ResNet-50 and VGG-16 require substantial computational resources, making them impractical for deployment on low-cost devices commonly used by farmers.

Another challenge lies in dataset imbalance, where certain crop classes dominate the dataset while others have limited samples, leading to biased model predictions. Therefore, the core problem addressed in this research is how to design a plant disease classification model that is both highly accurate and efficient enough to run on resource-constrained devices while maintaining robustness in real-world conditions.

The primary objective of this research is to develop a high-performance plant disease classification system that can be deployed in real-world agricultural settings. Specifically, the study aims to evaluate the performance of the EfficientNet family of models (B0–B7) and compare them with established architectures such as ResNet-50 and MobileNetV2. The research also focuses on optimizing model training using techniques such as transfer learning, data augmentation, class-weighted loss functions, and cosine learning rate scheduling to improve accuracy and generalization. Another key objective is to ensure real-time performance by converting the trained model into a TensorFlow Lite format with INT8 quantization for efficient mobile deployment.

Additionally, the system is designed to include a complete implementation pipeline consisting of a FastAPI backend and a React frontend, enabling users to upload images and receive disease predictions along with relevant information such as symptoms and treatment recommendations. Overall, the goal is to create a solution that is not only accurate but also practical and accessible for farmers.

The experimental results demonstrate that EfficientNet-B4 provides the best balance between accuracy and computational efficiency among all tested models. It achieved a top-1 classification accuracy of 95.9% on the test dataset, outperforming ResNet-50 by approximately 2.1 percentage points while using a comparable number of computational resources. Although larger models such as EfficientNet-B5, B6, and B7 achieved slightly higher accuracy, the improvement was marginal and came at the cost of significantly increased parameters and computational complexity, making them less suitable for mobile deployment.

After applying INT8 quantization, the model size was reduced from 76 MB to 22 MB with minimal loss in accuracy. The optimized TensorFlow Lite model achieved an inference time of approximately 22 milliseconds on a mid-range Android device, enabling real-time predictions. However, when tested on real-world field images collected from farms in Haryana, the model's accuracy dropped to 89.3%, highlighting the impact of domain shift between training and real-world data. Despite this reduction, the system still provides valuable assistance to farmers by delivering fast and reasonably accurate predictions.

This research successfully demonstrates that it is possible to design a plant disease classification system that is both accurate and practical for real-world deployment. EfficientNet-B4 emerged as the most suitable model, offering a strong trade-off between performance and efficiency. The integration of TensorFlow Lite optimization and mobile deployment ensures that the system can operate in real-time on affordable devices, making it accessible to farmers in rural areas.

However, the study also highlights a significant limitation in the form of reduced accuracy on real-world field images, emphasizing the need to address the domain shift problem. Future work should focus on collecting more diverse and representative field data, exploring semi-supervised learning approaches, and incorporating disease localization techniques to improve model reliability and usability. Expanding the dataset to include more staple crops such as wheat, rice, and soybean would further enhance the system's practical impact.

References

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Thank You